

Sahit Polineni

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EDUCATION

Bachelor of Science in Mechanical Engineering

University of Florida, Gainesville, FL

May 2027

EXPERIENCE

Terumo Neuro

R&D Engineering Intern

June 2026 – August 2026

Aliso Viejo, California

- **Designed and prototyped** a mechanical fixture to stabilize the fiber during pressure data collection, minimizing fiber movement and acoustic reverberation while enabling quick setup between test runs.
- Engineered a damage threshold and power transmission test methodology, applying a **DOE** to characterize and benchmark 5 fiber optic vendors, then developed a decision matrix to standardize fiber selection criteria.
- Designed, iterated, and implemented a CAD fixture to precisely position a blast shield along an optical path, utilizing **tolerancing analyses and measurement verification** to maintain optical alignment and protect an optical lens from blast-back across repeated high-power damage threshold tests.
- Completed the **hardware integration** of a 3-axis motion control gantry, integrating a power supply, Arduino mega, and stepper motor drivers to enable automated, repeatable positioning for spatially resolved pulse measurements.

Johnson & Johnson MedTech

Manufacturing Engineering Intern

June 2025 – August 2025

Jacksonville, Florida

- Led the redesign of a key feature in an FDA regulated manufacturing process, utilizing technician feedback to identify root causes and ultimately improve the longevity of equipment, **saving over \$55k** in repair and part costs.
- Collaborated with reliability, process, and manufacturing engineers to conduct design reviews, identify process inefficiencies, propose solutions that **reduced downtime by 30%** and increased efficiency across lines.
- Identified areas of improvement on the manufacturing line through root-cause analysis and DFMA principles, collaborating across cross-functional teams to design a solution to **save over \$100k** within a four-year period.
- **Developed detailed engineering drawings** in SolidWorks for external vendors, ensuring accuracy by verifying tolerances, materials specs, and manufacturability through collaboration with suppliers and manufacturing engineers.
- Utilized rapid prototyping to fit test redesigned assemblies and validate their integration into existing production line systems.

PROJECTS & RESEARCH

Dream Team Engineering

Operating Room Table Attachment Captain

August 2024 – Present

Gainesville, Florida

- Leading a team in the electromechanical design and manufacturing of a sterilizable, Raspberry Pi-based, surgical tool-tracking system, addressing up to **\$1M in annual surgical tool waste** while meeting operating room sterilization standards.
- Rapid prototyping device components using **3D printing** and planning for manufacturing through manual machining and CNC manufacturing processes to develop a product for a clinical setting.
- Utilized DOE principles to plan bench-top tests evaluating enclosure shape and clamp configuration on device performance.

Suture Pad and Surgical Design Team Engineer

- Designing a suturing training model to evaluate biases among physicians, using **SolidWorks** to create a casing to hold the accelerometer, and utilizing Arduino to integrate the sensor into the model.

Generational Relief in Prosthetics (GRiP)

Design Team Engineer

March 2024 – May 2025

Gainesville, Florida

- Applied collaborative problem-solving skills to conceptualize and design an ergonomic prosthetic device in Blender and SolidWorks, enabling individuals with limb differences to exercise with several types of free weights up to 35 pounds.
- Designed a casing for a robotic arm equipped with Myosensors to detect and process muscle twitches from the recipient's opposite arm and utilized servo motors to replicate this motion.

University of Florida College of Medicine

Research Assistant at The Center for Safety, Simulation, & Advanced Learning Technologies

March 2024 – February 2025

Gainesville, Florida

- Designed silicone molds for anatomical organs and **3D-printed models** for medical students to practice surgical procedures.
- Modeled and 3D-printed supplementary parts to improve the patient experience and physician efficiency in the hospital room.